Currency Depreciations, Credit Spreads, and Imported Inputs

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Introduction

How do economies respond to a depreciation?

Hotly debated topic

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- Relevant for:

  1. Central Banks: Inflation & **E**xchange **R**ate **P**ass **T**hrough (ERPT)
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- **Missing Link:** How does response of develop(ed/ing) countries differ?

This Paper

1. Empirical Analysis
   - Take data from multilaterals about CPI, FX, output & rates $\rightarrow$ estimate IRFs (In spirit of Jordà, 2005)
   - Estimate LPs with Emerging Market dummy interaction
   - *Depreciations have a greater impact in **E**merging **M**arkets than **A**dvanced **E**conomies*

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What could be behind these differences?

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What could be behind these differences?

2. Model with imperfect financial markets (Akinci, Queralto; 2024) and imported inputs
   - *Interaction* of financial and nonfinancial firms
   - ERPT and UIP
   - Simulation results replicate EM-AE gaps and match some volatilities

Empirical Analysis

Data

Macro Panel:

- 1990Q1-2023Q4
- 51 Countries - 30 AE & 21 EM
- 5,213 Observations
- Output, CPI, ER, Policy rates & Economic Classification

Details

Local Projections

1. Definitions
   - **Exchange rate shock:** $\hat{\varepsilon}_t^{FX} \equiv \Delta \ln FX_t - \mathbb{E}[\Delta \ln FX_t | \mathcal{F}_{t-1}]$
   - **Cumulative change:** $\Delta^h y_{i,t} \equiv y_{i,t+h} - y_{i,t-1}$.
     Example: $\Delta^3 \ln GDP_{Argentina,t} \equiv \ln GDP_{Argentina,t+3} - \ln GDP_{Argentina,t-1}$

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2. Projection

$$\Delta^h y_{i,t} = \gamma_t^h + \beta_1^h \hat{\varepsilon}_{i,t}^{FX} + \beta_2^h \left(\hat{\varepsilon}_{i,t}^{FX} \times EM_i\right) + \delta' \mathbf{X}_{i,t} + u_{i,t+h}^y \qquad \text{(PROJ)}$$

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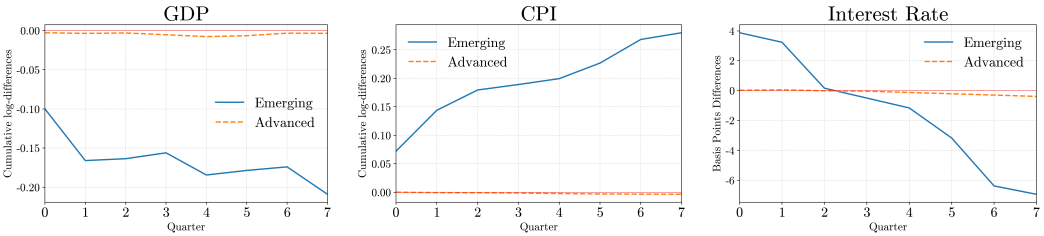
Assumption: $\mathbb{E}[\hat{\varepsilon}_t^{FX} u_{t+h}] = 0 \quad \forall h \geq 0$

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Results

After a 1% unexpected depreciation...

GDP falls more, CPI rises more, Rates react more in EM



Details

Robustness

- Estimate S-VAR as robustness check – asymptotically same as LP
  (Ólea, Plagborg-Møller, Qian, Wolf, 2025)

- Results consistent with LP: EM GDP falls more, CPI rises more & rates increase more than AE

Details

Model

Households, Banks and Government

Households: $[0, f]$ bankers, $(f, 1]$ workers & perfect household insurance with preferences:

$$\max \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t \Big(\log C_t - \frac{L_t^{1+\chi}}{1+\chi} \Big) \right]$$

Government: Fiscal authority is neutral & monetary authority targets nominal rate

$$R_{t+1}^f = \beta^{-1} \left(\pi_t^H \right)^{\gamma} \varepsilon_t^{Rf}$$

Bankers: Maximize continuation value s.t. limited commitment

$$V_{j,t} = \max \mathbb{E}_t \sum_{i=1}^{\infty} \Lambda_{t,t+i} (1-\theta_p) \theta_p^{i-1} N_{j,t+i} \quad \text{s.t.} \quad V_{j,t} \geq \Theta(x_{j,t}) Q_t A_{j,t}$$

with

$$N_{j,t} = R_t^K Q_{t-1} A_{j,t-1} + \left(R_t S_{t-1} - R_t^* S_t \right) D_{j,t-1}^* + R_t N_{j,t-1}$$

Asset prices and Financial Amplification

Banker's FOC

$$\sigma_t = \mathbb{E}_t \left[\Lambda_{t,t+1} \Omega_{t+1} R^K_{t+1} \right]$$
$$\zeta_t = \mathbb{E}_t \left[\Lambda_{t,t+1} \Omega_{t+1} \left(R_{t+1} - \frac{R^*_{t+1}}{S_t} S_{t+1} \right) \right]$$
$$\varrho_t = \mathbb{E}_t \left[\Lambda_{t,t+1} \Omega_{t+1} R_{t+1} \right]$$

Implications. Rate differentials pin down leverage and foreign liabilities share and capital intermediation:

- Higher ϕ_t, $x_t \Rightarrow$ more capital intermediated $\Rightarrow$ higher K_{t+1}, Y^{AH}_t

Non-Financial Firms

Capital producers: Install I_t s.t adjustment costs $\rightarrow Q_t$ (Tobin's Q)

Final Output: Sector $s \in [0,1]$ CES aggregator

Domestic goods firms: CES technology over domestic & imported inputs

- Input Allocation:

$$\frac{v_{s,t}}{m_{s,t}} = \frac{1-\omega}{\omega}\left(\frac{P_t^V}{e_t P_t^*}\right)^{-\eta}, \quad v_{s,t} = k_{s,t}^{\alpha} l_{s,t}^{1-\alpha}$$

- Optimal Price:

$$p_{s,t}^H = \mu \frac{PPI_t}{\Gamma_t}$$

RoW and Aggregation

Rest of the world

$$R_t^* = R_t^W \exp\left\{\kappa\left(\frac{D_t^*}{P_t^*} - \bar{D}^*\right)\right\},$$

$$C_t^{H*} = \left(\frac{P_t^H}{e_t P_t^*}\right)^{-\varepsilon^*} Y_t^*.$$

Market-clearing conditions

$$A_t = K_{t+1} = (1-\delta)K_t + I_t$$

$$Y_t^{AH} = C_t^H + C_t^{H*} + I_t + \frac{\psi}{2}\left(\frac{I_t}{K_t} - \delta\right)^2 K_t$$

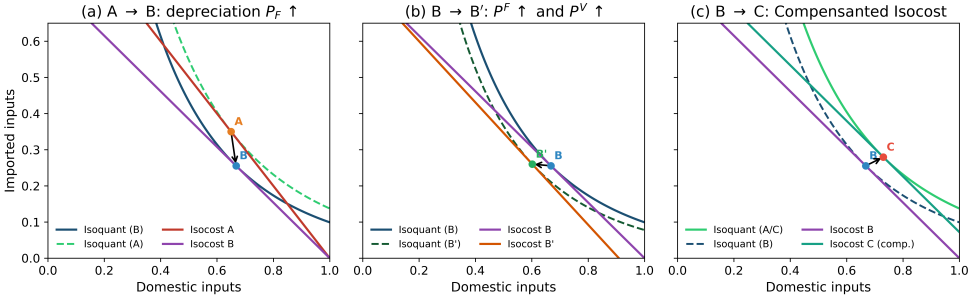
$$S_t\left(D_t^* - R_t^* D_{t-1}^*\right) = \frac{P_t^* e_t}{P_t}\left(C_t^F + M_t\right) - \frac{P_t^H}{P_t} C_t^{H*}$$

Model Results

Key Mechanisms:

1. ERPT:

$$\frac{d \log P_t^H}{d \log e_t} = \frac{(1-\omega)(P^V)^{1-\eta}}{PPI^{1-\eta}} \frac{\partial \ln P^V}{\partial \ln e} + \frac{\omega (P^* e)^{1-\eta}}{PPI^{1-\eta}}$$

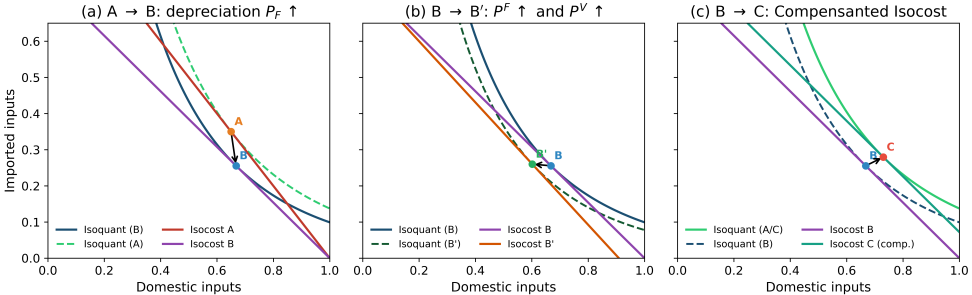


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2. UIP: $R_t^W \uparrow \Rightarrow$ expected appreciation path $\Rightarrow$ real depreciation today $S_t \uparrow$

Parameters

- **Key Parameter**: $\text{Corr}(\varepsilon_\Gamma, \varepsilon_{R^W})$: < 0 for EM & 0 for AE
  (Neumeyer, Perri, 2005)

- Others standard for *both* economies (Akinci, Queralto, 2024) Details

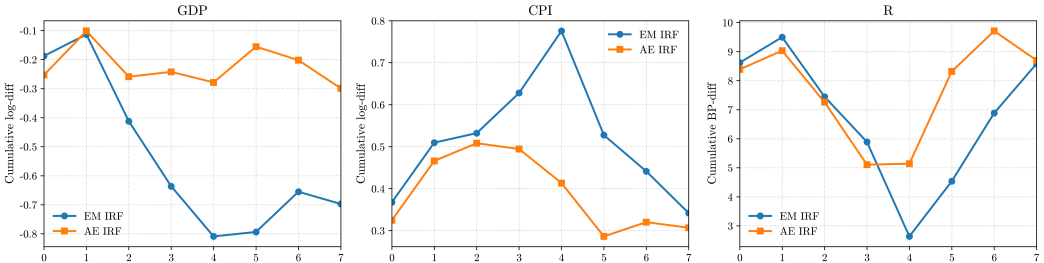
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   - Simulate 2,000 quarters & keep last 40
   - Run (PROJ) with simulated data

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2. Model vs Data Moments

	AE		EM		Bigger?		
Moments	Data	Model	Data	Model	Data	Model	Match
$\sigma(\Delta\%\ FX)$	3.8%	2.3%	4.3%	2.8%	EM	EM	✓
$\sigma(\Delta\%\ RER)$	3.8%	1.3%	4.1%	1.5%	EM	EM	✓
$\sigma(\Delta\% CPI)$	0.9%	1.2%	2.3%	1.3%	EM	EM	✓
$\sigma(\Delta\%$ GDP)	2.6%	2.9%	4.0%	2.3%	EM	AE	X
$\sigma(R)$	2.9%	0.8%	11.0%	0.6%	EM	AE	X

Conclusion

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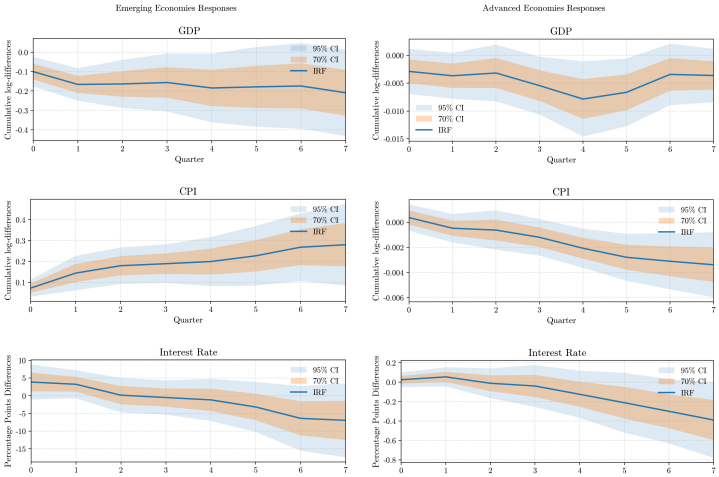
1. **Main Feature:** Replicate qualitative EM-AE differences with joint distribution of shocks
    - Suggests depreciations coincide with productivity weakness

Conclusion

1. **Main Feature:** Replicate qualitative EM-AE differences with joint distribution of shocks
   - Suggests depreciations coincide with productivity weakness
2. **Main Limitation:** No endogenous mechanism
   - Cannot disintangle whether EM fragility stems from banks, production, or shock structure

Appendix

Local Projections



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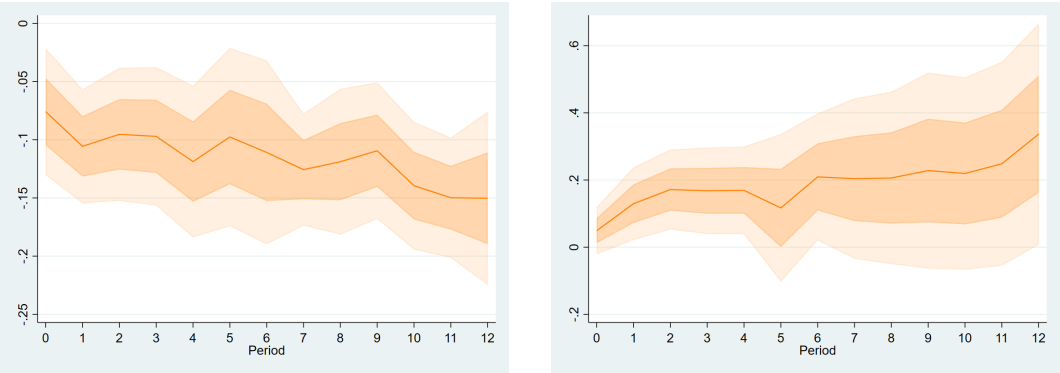


Figure: Cumulative IRF of GDP and CPI amid a 1 % REER shock

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Macro Data

Macro Panel:

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- 51 Countries - 30 AE & 21 EM
- 5,213 Observations
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Variable	Freq	Source	Transformation
Output SA Deflator	Q	IMF	LN
CPI	Q	IMF	LN
Exchange Rate to USD	Q	BIS	LN
Policy Rates (Annualized)	Q	BIS/FRED	BP (100bp = 1%)
Economic Classification	Y	World Bank	N/A

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Financial Industry Regressions

Micro Panel:

- 1986Q1-2025Q1
- 40 Countries - 25 AE & 15 EM
- 5,128 entities
- 294,504 observations
- Index price, Stock price, CPI, ER, Policy rates & Economic Classification

Variable	Freq	Source	Transformation
Index Price	D	Compustat	Avg real Q return
Stock Price	D	Compustat	Avg real Q return
Exchange Rate to USD	Q	BIS	Avg real %change
Policy Rates (Annualized)	Q	BIS/FRED	Net Q real rate
Economic Classification	Y	World Bank	N/A

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Financial Industry Regressions

- Panel FE regression (bank–quarter):

$$R_{i,t} = \alpha_i + \beta\, \Delta\%FX_{c(i),t} + \delta(EM_{c(i)} \cdot \Delta\%FX_{c(i),t}) + \gamma_0 R^{mkt}_{c(i),t} + \gamma_1 r^{real}_{c(i),t} + \gamma_2 r^{*,real}_{t} + \varepsilon_{i,t}$$

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Financial Industry Regressions

	Coefficient	Coefficient (Excludes Japan & UK)
Real Return Index	0.430***	0.365***
	(0.043)	(0.042)
Real Depreciation	-0.045***	-0.066***
	(0.014)	(0.013)
Real Depreciation × EM	-0.403***	-0.416***
	(0.047)	(0.047)
Real Domestic Interest Rate	-0.003**	-0.003***
	(0.001)	(0.001)
Real FED Interest Rate	-0.278***	-0.118**
	(0.050)	(0.053)
Constant	0.024***	0.024***
	(0.001)	(0.001)
Observations	288,580	200,889
R^2	0.064	0.061
Adjusted R^2	0.047	0.042
Fixed effects	Country-Firm	Country-Firm

***, ** denote significance at 1%, 5% level respectively.

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SVAR Results

	Emerging Market Economies			Small-open Advanced Economies		
Horizon	Δr	$\Delta \ln$ CPI (%)	$\Delta \ln$ GDP (%)	Δr	$\Delta \ln$ CPI (%)	$\Delta \ln$ GDP (%)
2 quarters	-2.64	0.25*	-0.42	-6.23*	-0.02	-0.19
	(21.72)	(0.19)	(1.66)	(5.46)	(0.12)	(0.40)
4 quarters	5.52	0.36	-0.47	-6.98	-0.04	-0.27
	(40.42)	(0.39)	(1.07)	(11.09)	(0.23)	(0.72)
8 quarters	5.96	0.44	-0.56	-9.16	-0.07	-0.23
	(59.54)	(0.60)	(1.66)	(16.47)	(0.34)	(1.03)

Cumulative IRFs to a 10% exchange-rate depreciation. CPI and GDP are expressed in percentages, whereas the interest rate is in basis points. Bootstrap standard errors in parentheses. Entries show coefficients with significance (***, **, *) for 5%, 10%, 30% levels.

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Parameter Values

Parameter	Symbol	Value (AE)	Value (EM)	Source/Target
Household preferences				
Discount factor	β	0.9925		Annualized SS $R \approx 3\%$
Labour disutility	χ	2		SS labor $\approx 30\%$
Preference weight on foreign goods	ω	0.35		Benchmark value
Substitution elasticity home/foreign goods	η	1.3		Standard in literature
Production				
Sector elasticity of substitution	ϵ	11		Markup of 10%
Capital share	α	0.33		Standard in literature
Capital depreciation rate	δ	0.025		Standard in literature
Adjustment cost sensitivity	ψ_I	10		AQ (2023)
Financial intermediaries				
Survival rate	θ_p	0.95	0.95	AQ (2023)
Divertability parameter 1	$\theta_{\mathbf{0}}$	0.52	0.52	Derived from SS targets
Divertability parameter 2	$\theta_{\mathbf{1}}$	0.41	0.41	Derived from SS targets
Home bias in bank funding	$\theta_{\mathbf{2}}$	2.06	2.06	Derived from SS targets
Transfer rate to new entrants	ξ	0.067	0.067	AQ (2023)
Monetary policy rule				
Response to inflation	γ	1.5		Standard in literature
Rest of the world				
Foreign demand elasticity	ϵ^*	0.3		Small value
Debt elasticity of foreign interest rate	κ	0.001		Small value
Exogenous processes				
Technology AR parameter	ρ_Γ	0.9	0.9	Persistent process
Foreign interest-rate AR parameter	ρ_{R^W}	0.5	0.5	Moderate persistence
Foreign output AR parameter	ρ_{Y^*}	0.5	0.5	Moderate persistence
Foreign price-level AR parameter	ρ_{P^*}	0.5	0.5	Moderate persistence
Taylor-rule shock AR parameter	$\rho_{\varepsilon_{R^f}}$	0.5	0.5	Moderate persistence

Go back